Phone Recorder user guide

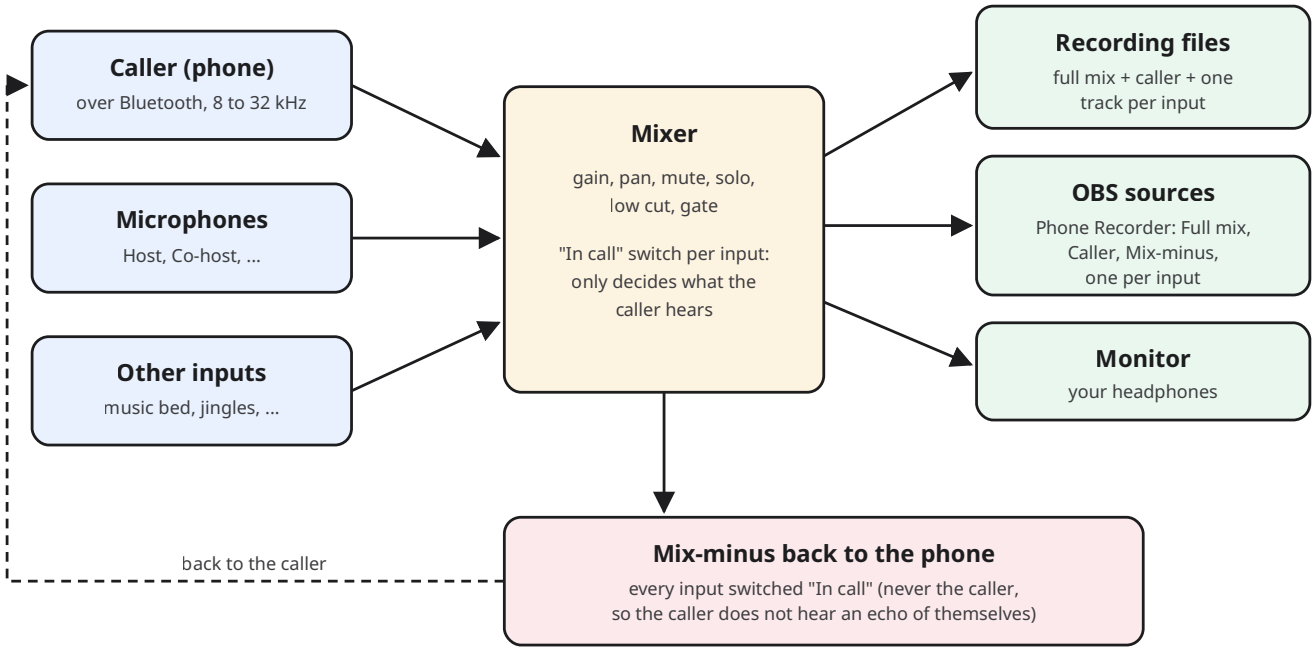
Phone Recorder records phone calls over Bluetooth for podcasts. Your computer connects to the phone the way a car kit does, so any call on the phone comes through the app: normal calls and calling apps such as WhatsApp or Signal. The caller and every microphone are recorded on separate tracks, and all of them are also available to OBS.

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Using OBS? See the separate OBS guide.

How audio flows through Phone Recorder



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1. What you need

- **A Linux computer** with PipeWire (standard on current distributions).
- **A Bluetooth adapter.** A separate USB dongle is best, because the app takes full control of the adapter it uses: other Bluetooth devices on that adapter (headphones, mouse, keyboard) disconnect while the app runs. A cheap, well-supported choice is the TP-Link UB500. The built-in adapter works too. See the hardware table in the README.
- **Headphones** for monitoring. With speakers the caller may hear an echo (there is an echo canceller for that case, see the mixer).
- **One-time setup** (asks for your password): This lets your user drive USB Bluetooth adapters without root rights. Building and installing the app is described in BUILD.md.

```
scripts/install-udev-rule.sh
```

Which phones work

Any phone that works with a car kit or a Bluetooth headset: Android phones, iPhones and most other mobile phones (Bluetooth Hands-Free Profile). The contact list needs the phone to share its address book (Phone Book Access Profile), which Android phones and iPhones do after you allow it; some older phones cannot, and then the app simply works without contacts.

About sound quality

Bluetooth calls are limited by the phone link itself, no matter what the app records at:

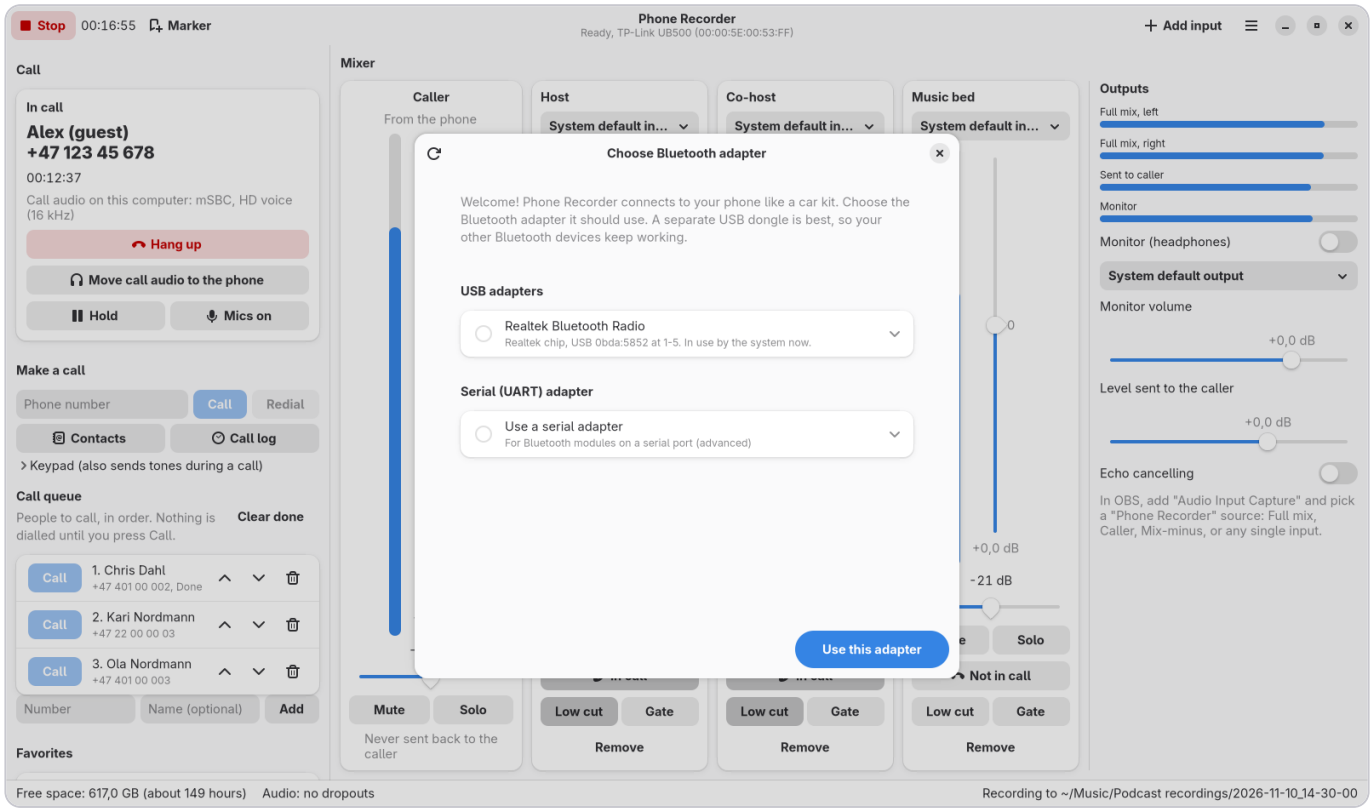
Shown during the call	Sample rate	Typical phones
CVSD, narrowband	8 kHz	old phones or adapters
mSBC, HD voice	16 kHz	most phones from the last decade
LC3-SWB, super wideband	32 kHz	newer phones

The app picks the best one the phone and the adapter both support. Your own microphones are recorded at full quality (48 kHz, 32-bit).

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2. First start: choose the Bluetooth adapter

On the first start the app asks which Bluetooth adapter to use.



1. Pick your adapter. Each entry shows the chip, the USB ID and whether the system uses it right now.
2. Optional: open the entry to choose the **Mode**:
   - **Direct USB (recommended):** the app talks to the adapter itself and loads its firmware.

- **Kernel driver, raw access:** for chips whose firmware only the Linux kernel can load (MediaTek, Intel AX210 and newer, Qualcomm). Needs the helper: run `scripts/setup-userchan-helper.sh` once.

3. Choose **Use this adapter**. If the system is using it, the app first warns that other Bluetooth devices on it will disconnect. The app then restarts once.

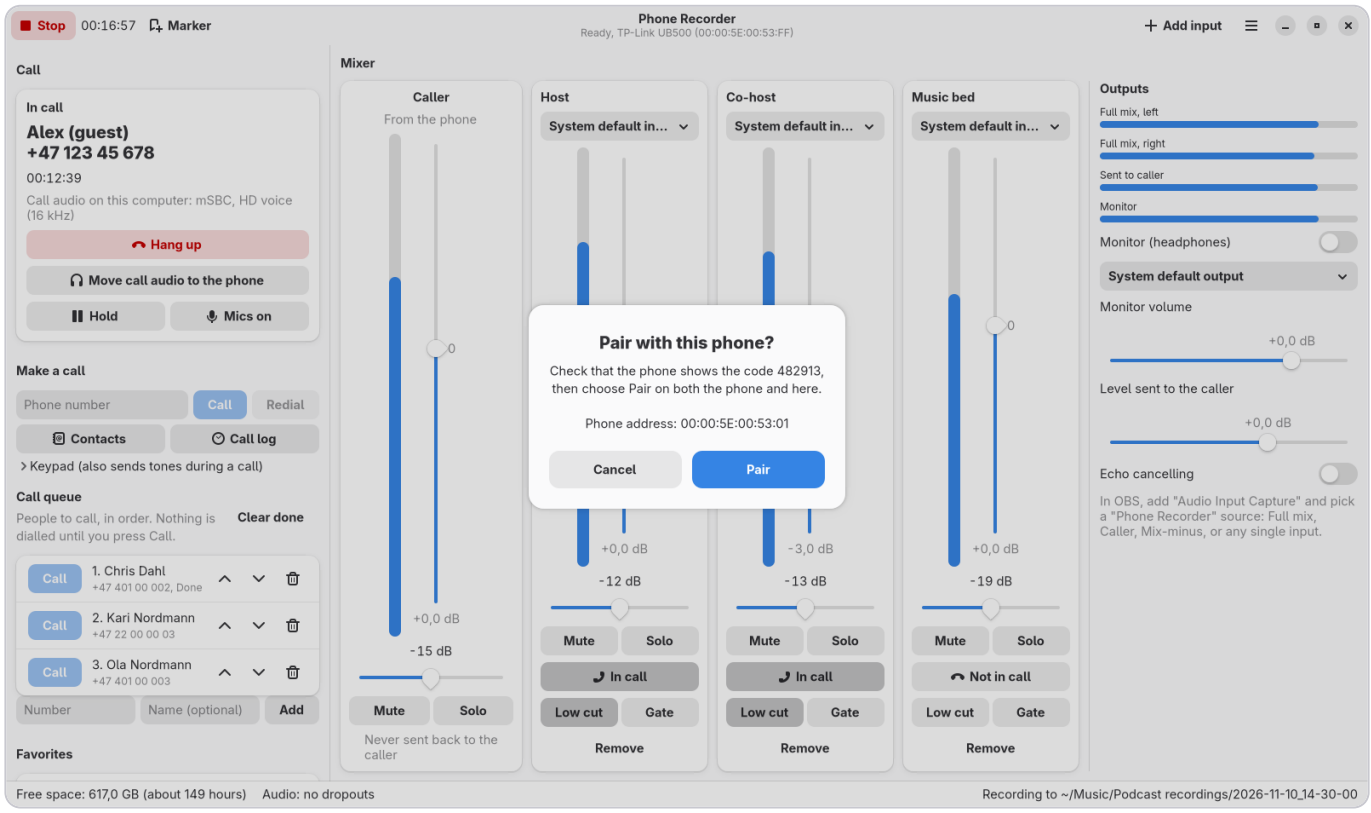
While the app runs, that adapter belongs to the app. When you close the app, it is given back to the system, also if the app should ever crash.

To switch adapters later: **Menu (☰), Choose Bluetooth adapter**.

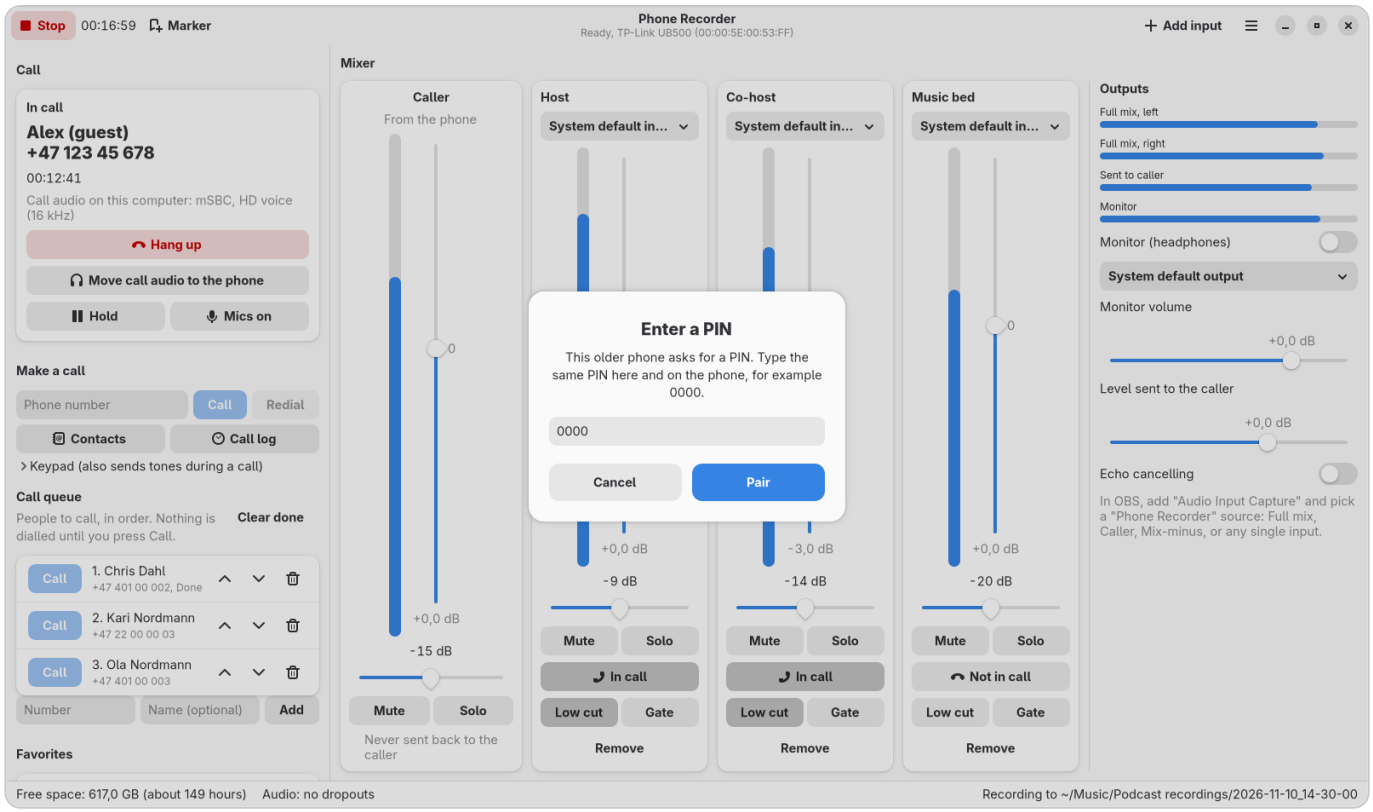
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3. Pair your phone

1. In the **Phone** section, turn on **Pairing mode**.
2. On the phone, open the Bluetooth settings and choose the app's name (by default "Phone Recorder", see Bluetooth name).
3. The app shows a code. Check that the phone shows **the same code**, then choose **Pair** on both the phone and the computer.



Older phones may ask for a PIN instead. Type the same PIN on both sides, for example 0000:



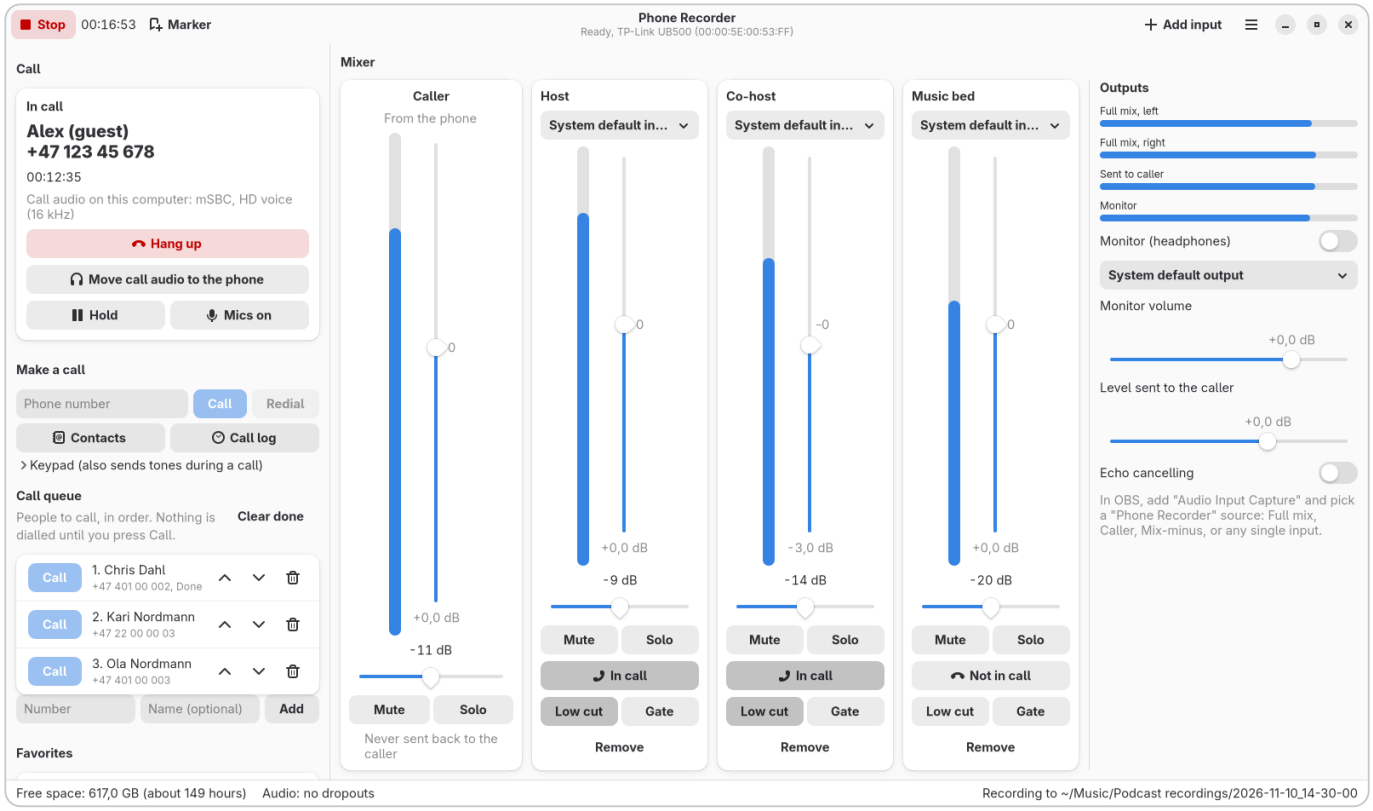
After pairing, the phone appears under **Paired phones**. Next time it connects by itself when the app starts (or use **Connect** in the list). **Forget** (the bin icon) removes a phone; it must then be paired again.

Bluetooth name

The name phones see can be changed in the **Phone** section: select the **Bluetooth name** field, type a new name and press Enter. It changes right away. Phones that are already paired may show the old name until they connect again.

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4. The main window



- **Header bar: Record** / **Stop** with the recording time, **Marker**, **Add input** and the main menu (☰). The subtitle shows the Bluetooth status.
- **Left side:** the current call, **Make a call**, the **Phone** section (connection, battery, signal, Bluetooth name, pairing mode) and **Paired phones**.
- **Right side:** the **Mixer** with one strip per source and the **Outputs** section.
- **Status line at the bottom:** free disk space with the recording time left, an audio health indicator ("no dropouts" or a count) and where the current recording is saved.

5. Calls

- **Incoming call:** the call card shows the caller's name or number and **Answer** and **Decline** (Ctrl+Shift+A, Ctrl+Shift+H). If the window is in the background, a desktop notification with an Answer button appears.
- **Make a call:** type the number under **Make a call** and press Enter or **Call**. **Redial** calls the last number again. The **Keypad** sends tones during a call (for menus like "press 1").
- **During the call:** the card shows the call time and the codec, for example "mSBC, HD voice (16 kHz)". **Hang up** ends the call.

- **Call audio on the phone or the computer: Move call audio to the phone** hands the audio back to the handset (the call continues); **Move call audio to the computer** brings it back (Ctrl+Shift+T). The app brings the audio to the computer by itself when a call starts.
- **Call waiting:** if a second call comes in during a call, the card shows "Call waiting" with the new caller's name or number and, below it, the call you are in. **Answer** puts the current call on hold and takes the new one; **Decline** rejects only the new call.

Closing the app during a call does not end the call; it continues on the phone.

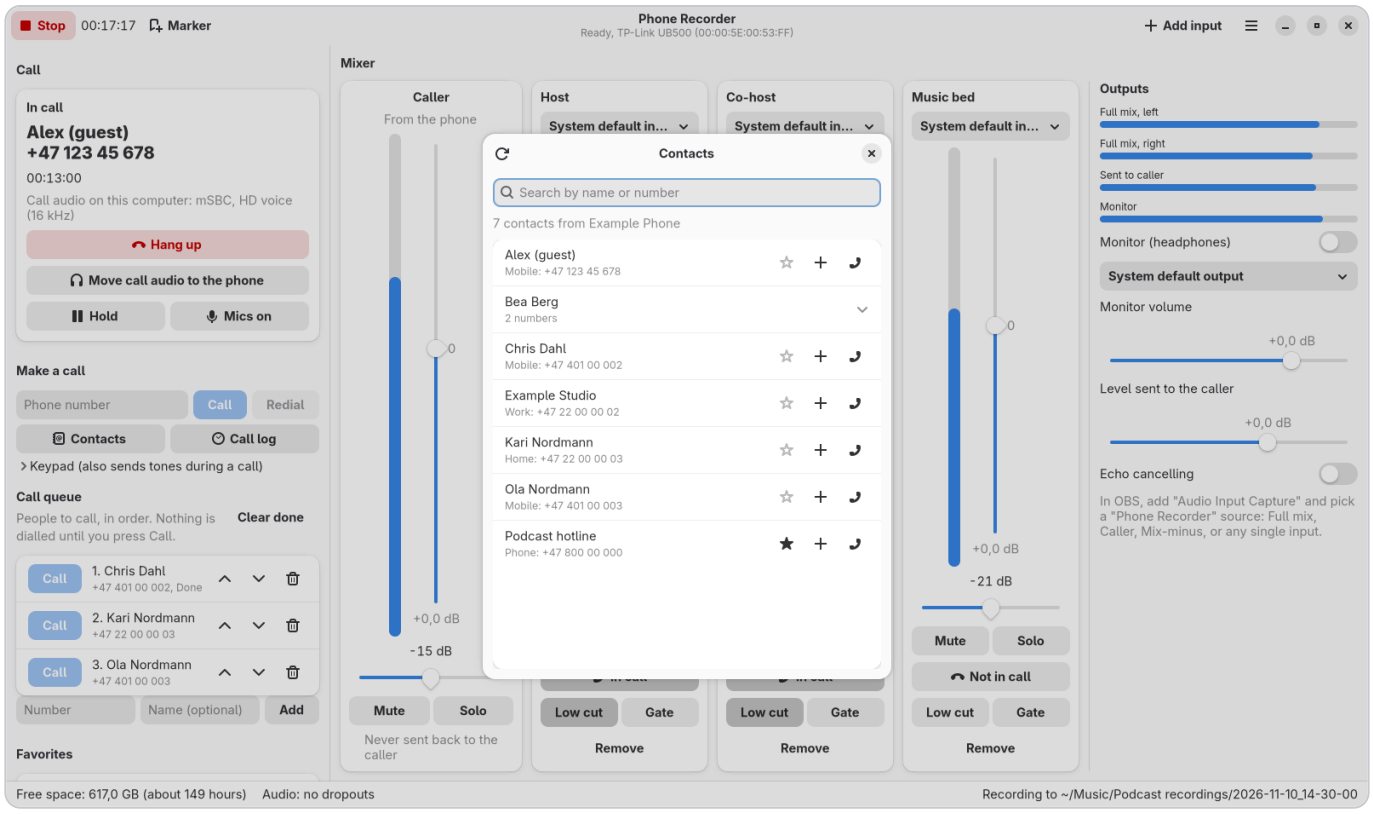
Hold and microphones off

During a call, two more buttons appear in the call card:

- **Hold** (Ctrl+Shift+P) puts the call on hold; the card says "On hold". **Resume** takes it back. This uses the phone's own hold, so the caller hears what the phone or network plays for calls on hold. A few phones cannot hold a single call; they then say so.
- **Mics on / Mics off** (Ctrl+Shift+M) decides whether the caller hears your microphones at all, for example to talk to a co-host off air. OBS, the recording and your headphones keep the microphones either way. It is always on again when the next call starts. To switch single inputs, use their **In call** buttons in the mixer.

Contacts

When a phone connects, the app downloads its address book, so you can call people by name and see who is calling even when the phone only sends a number. The contacts are kept only on this computer, in a file only your user can read, and are deleted when you **Forget** the phone.



- Open it with **Contacts** under **Make a call**, or Ctrl+K.
- Type in the search field to find a name or a part of a number.
- Click the call button next to a number to call it. Contacts with several numbers open to show them all.
- The refresh button (top left) downloads the address book again.

The phone must allow this the first time:

- **Android:** the phone asks whether to share contacts (and call history) with "Phone Recorder": choose **Allow**. If you missed the question, open the phone's Bluetooth settings, tap the gear next to Phone Recorder and turn on **Contacts** (sometimes called **Contact sharing**).
- **iPhone:** open **Settings, Bluetooth**, tap **(i)** next to Phone Recorder and turn on **Sync Contacts**. Then choose refresh in the Contacts window.

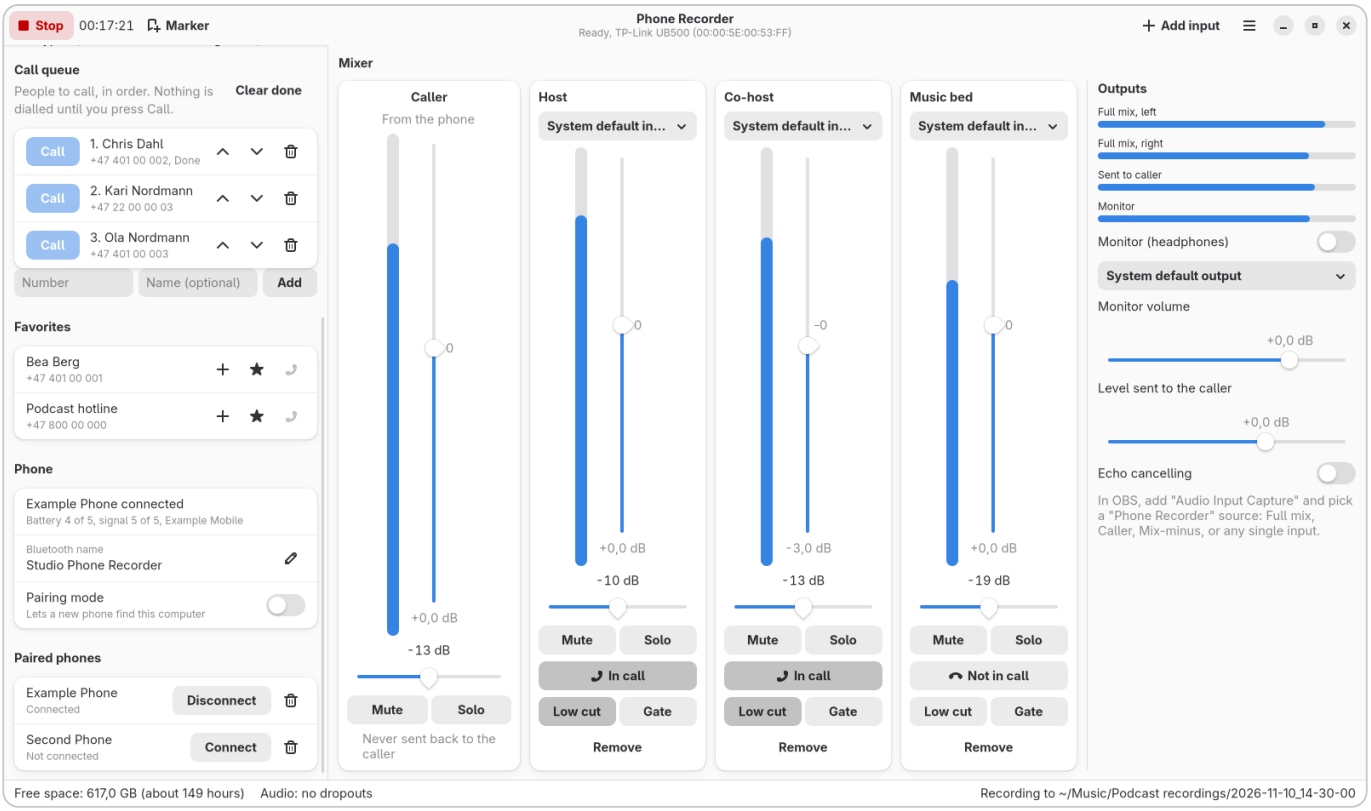
To stop downloading contacts, turn off **Download the phone's contacts** in Preferences, Bluetooth.

Favorites

The people you call most. Click the star next to a number in **Contacts** or the **Call log** to add or remove a favorite. Favorites are listed in the main window, each with a call button and a button to put them in the call queue.

Call queue

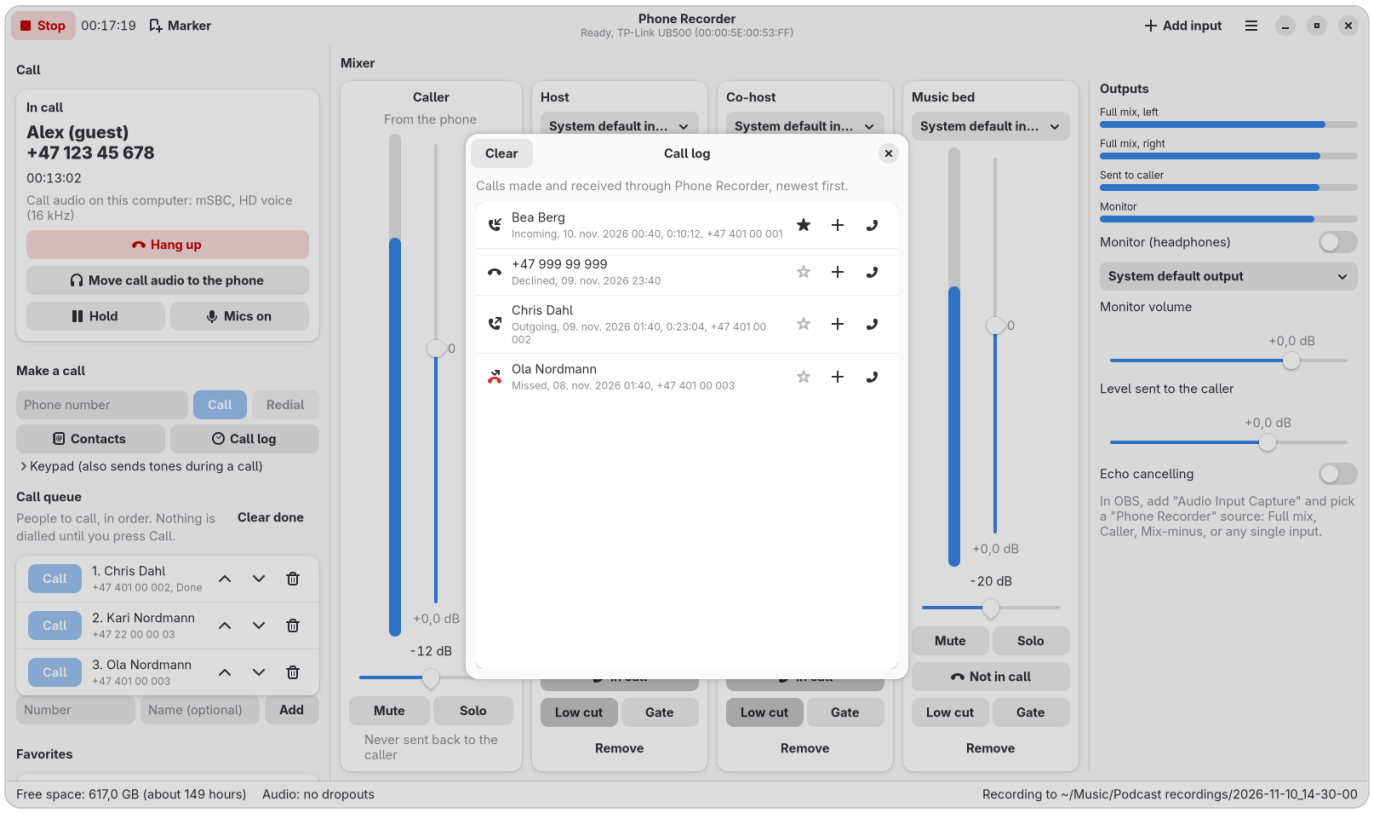
A list of people to call, in order, for example the guests of today's episode. **The queue never dials by itself**: you start each call with its own **Call** button, when you are ready.



- Add someone: type the number (and a name if you like) in the queue's fields and press **Add**, or use the **+** button next to a contact, a favorite or a call log entry.
- Change the order with the up and down arrows; remove an entry with the bin.
- When a call started from the queue is answered, the entry is marked **Done**. **Clear done** removes those.
- The queue is kept when you close the app.

Call log

Call log (under **Make a call**, or Ctrl+L) lists the calls made and received through Phone Recorder, newest first: incoming, outgoing, missed and declined, with date, time and how long you talked. From each entry you can call back, star it as a favorite or add it to the queue. **Clear** deletes the log from this computer (the phone's own call history is not touched).



6. The mixer

Each source has a strip. The **Caller** strip is always first.

Control	What it does
Name (click to rename)	Used for the recording file and the OBS source name.
Input device	Which microphone or input the strip uses. "System default input" follows your desktop setting.
Level meter and number	Peak level. Aim for about -12 dB when speaking. "CLIP" means too loud: turn the volume down.
Volume slider	Gain of the strip, -30 to +24 dB.
Left-right slider	Position in the stereo mix.
Mute	Takes the strip out of the mix and the call. Its own OBS source and recording track keep the sound.
Solo	Listen only to soloed strips on the monitor. Changes nothing in the recording, the call or OBS.
In call / Not in call	Whether the caller hears this input. **It never removes the input from OBS or the recording.**
Low cut	Removes rumble and handling noise below 80 Hz.
Gate	Turns the strip down when nobody speaks into it.
Remove	Removes the input (not possible while recording).

The caller is **never** sent back to the phone, so the caller never hears an echo of themselves. This is called a mix-minus.

Add input (Ctrl+I) adds another microphone or input, for example a music player you want on its own track.

Outputs

- **Meters:** full mix (left and right), what is sent to the caller, and the monitor.
- **Monitor (headphones):** turn on, choose the output device and set its volume. The monitor plays the full mix, or only the soloed strips.
- **Level sent to the caller:** overall volume of what the caller hears.
- **Echo cancelling:** only needed when the caller is played on speakers.

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7. Recording

- Press **Record** (Ctrl+R) to start and **Stop** to finish. A message tells you where the recording was saved, with an **Open folder** button.
- Each recording gets its own folder, named by date and time, for example `~/Music/Podcast recordings/2026-11-10_14-30-00/`.
- Inside are one file per source, all exactly in step:

```
01-full-mix.wav        the stereo mix
02-caller.wav          the caller
03-host.wav            each input, named after its strip
04-co-host.wav
markers-audacity.txt   markers for Audacity (File, Import, Labels)
chapters.txt           markers as chapter times
```

- **Markers** (Ctrl+M or F9) mark a point in the recording, for example a good quote. They are stored in the WAV files as cue points, as an Audacity label file and as a chapter list.
- **Automatic recording:** in Preferences, recording can start by itself when a call starts and stop when it ends.
- Recordings are safe if the computer crashes or loses power: the files stay readable up to the last moment.
- The status line warns when less than 15 minutes of disk space are left.

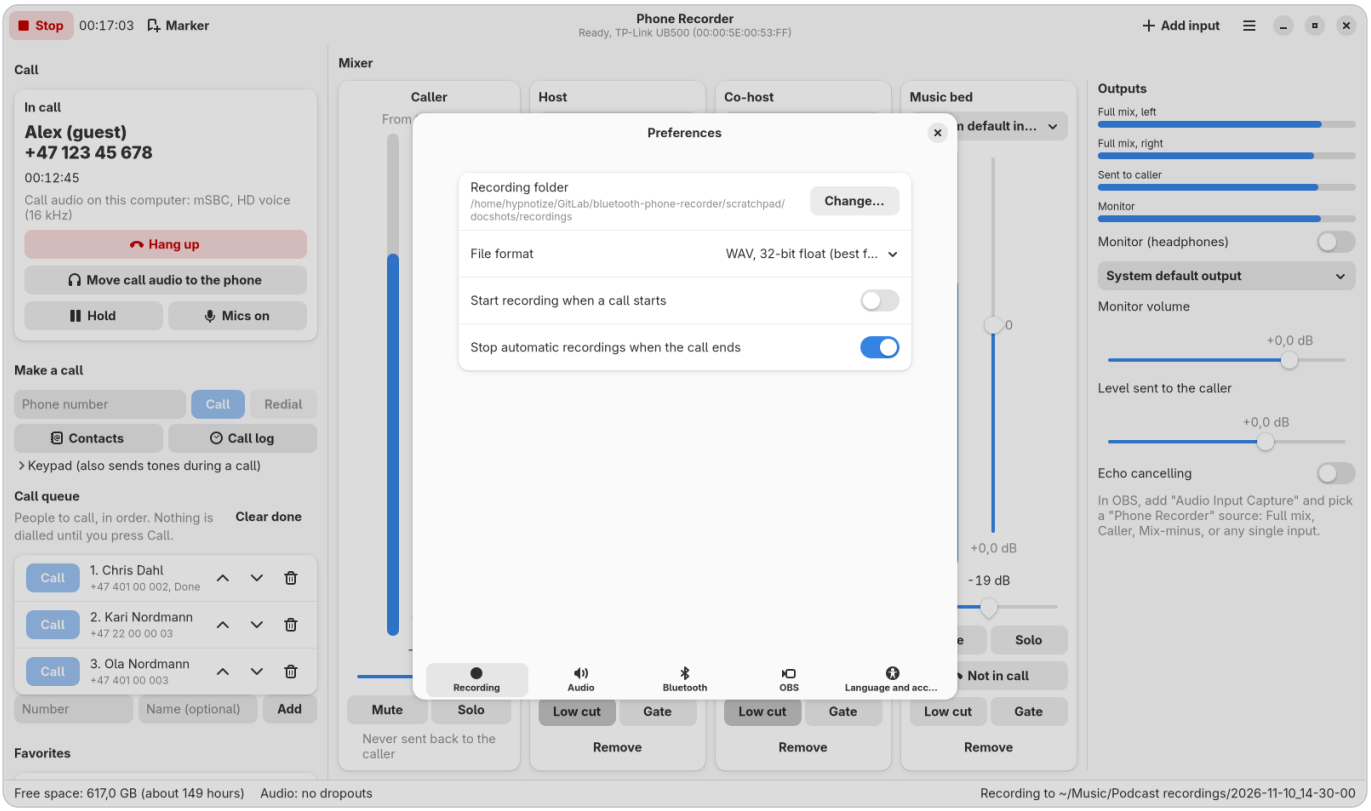
Inputs cannot be added or removed while recording.

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8. Preferences

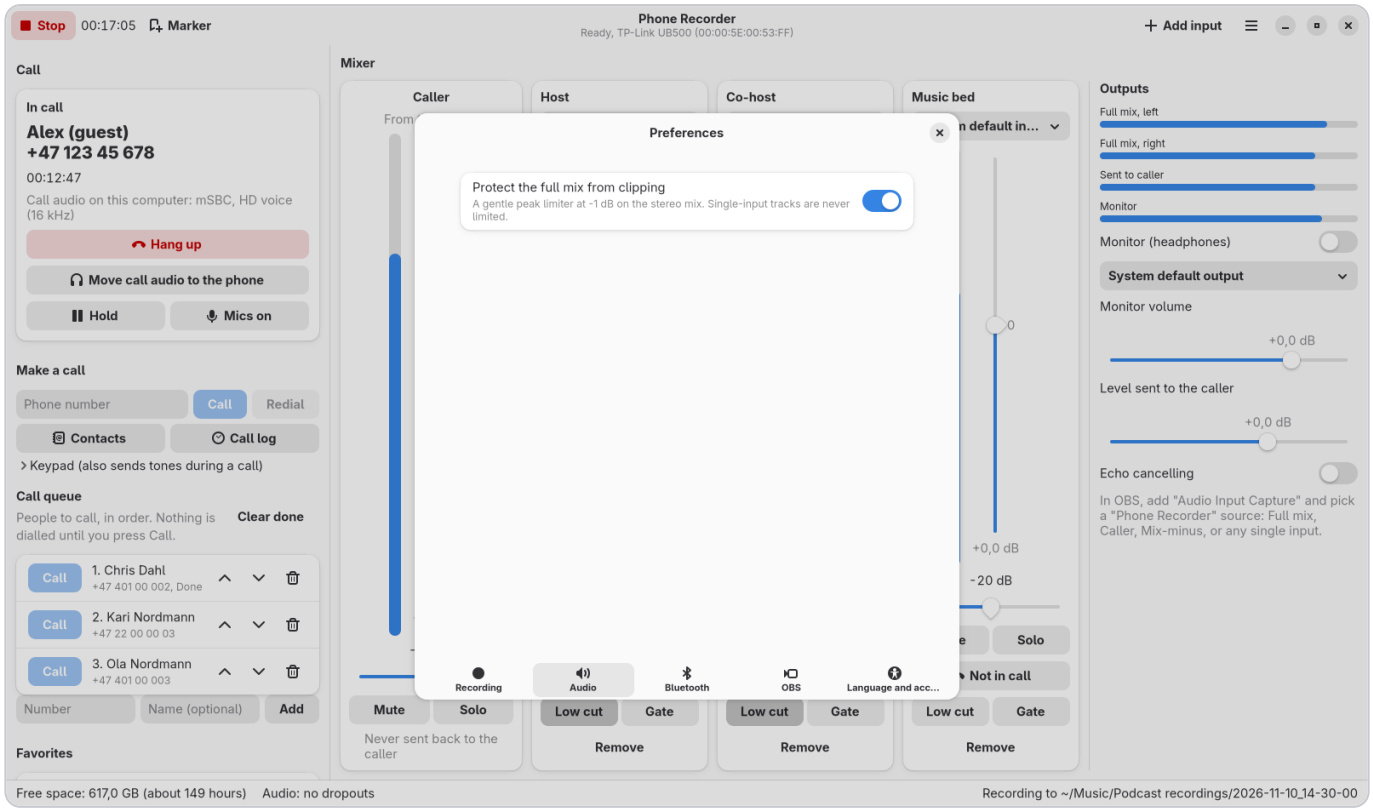
Open with **Menu (☰), Preferences** or Ctrl+comma.

Recording



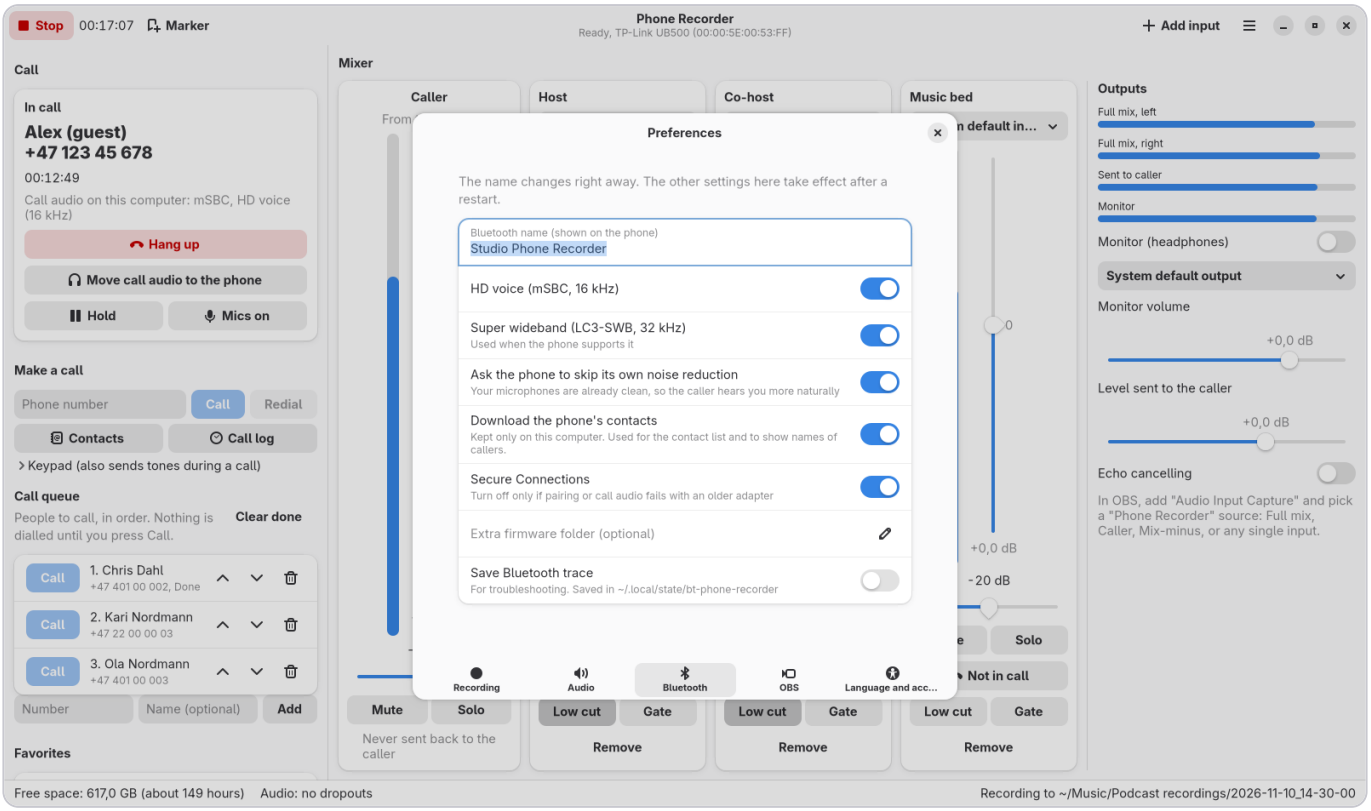
- **Recording folder.**
- **File format:** WAV 32-bit float (best for editing, default), WAV 24-bit, FLAC 24-bit (lossless, about half the size) or Opus (small, for sharing).
- **Start recording when a call starts** and **Stop automatic recordings when the call ends.**

Audio



- **Protect the full mix from clipping:** a gentle limiter on the stereo mix. Single-input tracks are never limited.

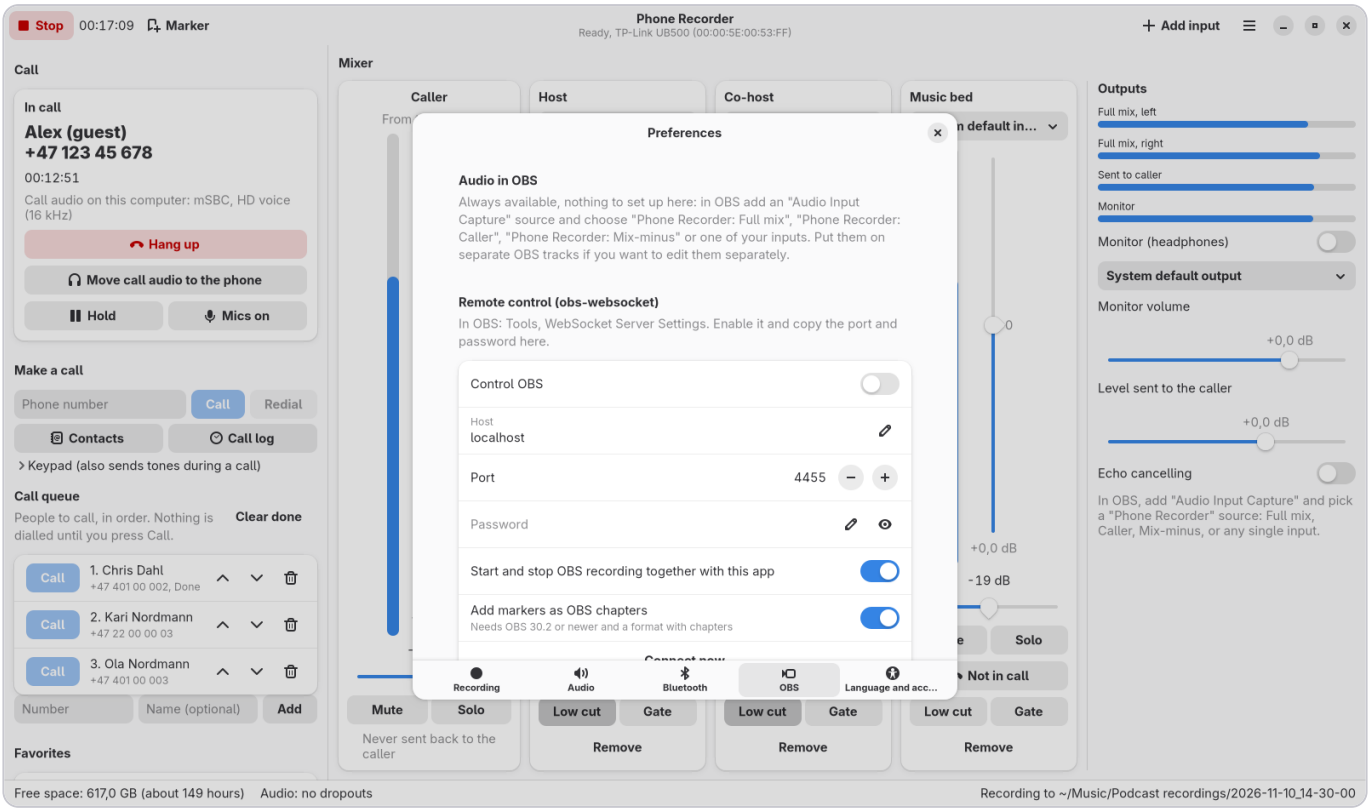
Bluetooth



- **Bluetooth name** (changes right away).
- **HD voice** and **Super wideband:** allow the better codecs. Leave on.
- **Ask the phone to skip its own noise reduction:** your microphones are already clean, so the caller hears you more naturally.
- **Download the phone's contacts:** see Contacts.
- **Secure Connections:** turn off only if pairing or call audio fails with an older adapter.
- **Extra firmware folder:** for adapters whose firmware is not in the system.
- **Save Bluetooth trace:** for troubleshooting (see below).

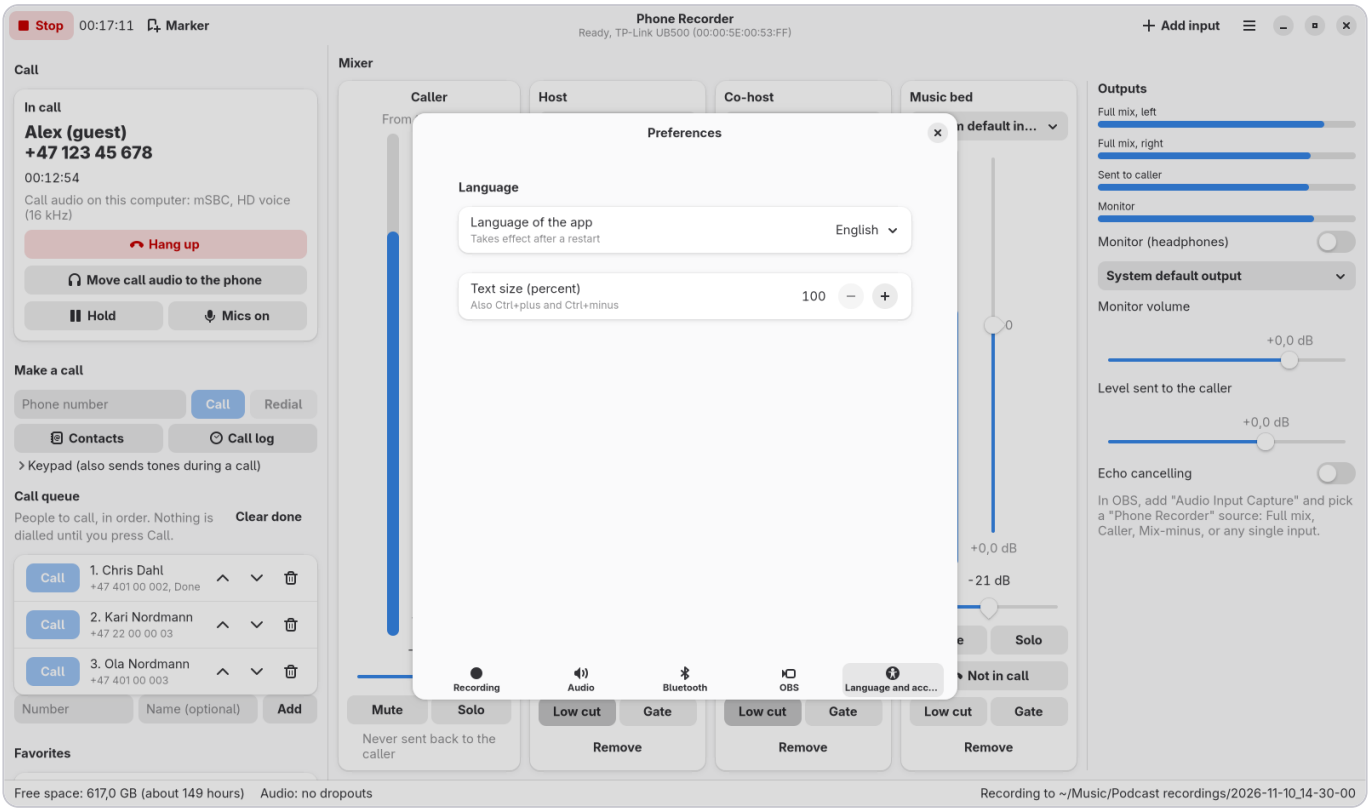
Apart from the name, these take effect after a restart.

OBS



See the OBS guide.

Language and accessibility

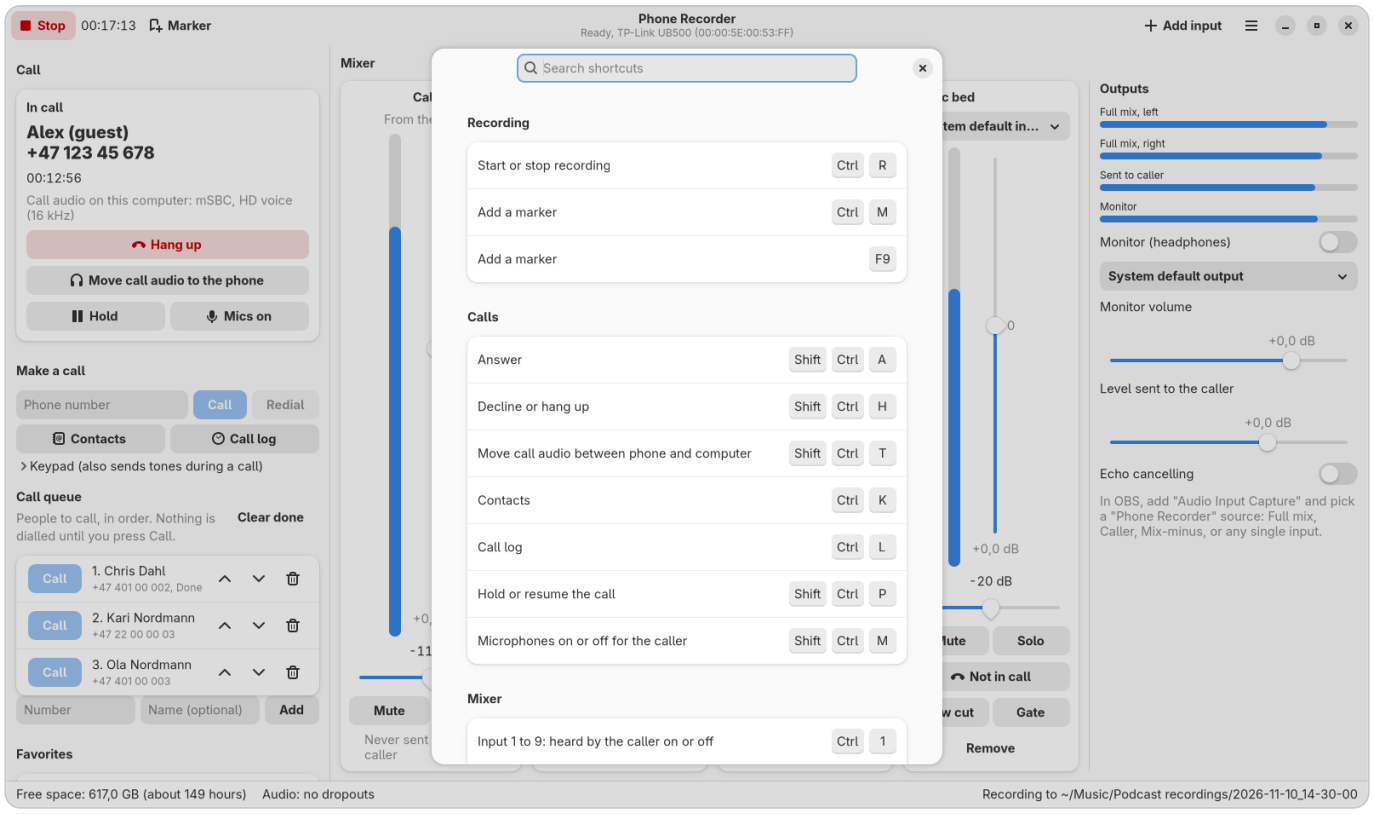


- **Language of the app:** System default, English or Norsk bokmål. Takes effect after a restart; the message that appears has a **Restart now** button.
- **Text size:** 100 to 200 percent, on top of your desktop setting.

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9. Keyboard shortcuts

Press F1 (or Ctrl+?) to see them in the app.


Stop
00:17:13
Marker
Phone Recorder
Ready, TP-Link UB500 (00:00:5E:00:53:FF)
Add input
Call
In call
Alex (guest)
+47 123 45 678
00:12:56
Call audio on this computer: mSBC, HD voice (16 kHz)
Hang up
Move call audio to the phone
Hold
Mics on
Make a call
Phone number
Call
Redial
Contacts
Call log
Keypad (also sends tones during a call)
Call queue
People to call, in order. Nothing is dialled until you press Call.
Clear done
Call
1. Chris Dahl
+47 401 00 002, Done
Call
2. Kari Nordmann
+47 22 00 00 03
Call
3. Ola Nordmann
+47 401 00 003
Number
Name (optional)
Add
Favorites
Mixer
From the
Mute
Never sent to caller
Search shortcuts
Recording
Start or stop recording
Ctrl R
Add a marker
Ctrl M
Add a marker
F9
Calls
Answer
Shift Ctrl A
Decline or hang up
Shift Ctrl H
Move call audio between phone and computer
Shift Ctrl T
Contacts
Ctrl K
Call log
Ctrl L
Hold or resume the call
Shift Ctrl P
Microphones on or off for the caller
Shift Ctrl M
Mixer
Input 1 to 9: heard by the caller on or off
Ctrl 1
c bed
tem default in...
+0,0 dB
-20 dB
Mute
Solo
Not in call
w cut
Gate
Remove
Outputs
Full mix, left
Full mix, right
Sent to caller
Monitor
Monitor (headphones)
System default output
Monitor volume
+0,0 dB
Level sent to the caller
+0,0 dB
Echo cancelling
In OBS, add "Audio Input Capture" and pick a "Phone Recorder" source: Full mix, Caller, Mix-minus, or any single input.
Free space: 617,0 GB (about 149 hours)
Audio: no dropouts
Recording to ~/Music/Podcast recordings/2026-11-10_14-30-00

Keys	Action
Ctrl+R	Start or stop recording
Ctrl+M or F9	Add a marker
Ctrl+Shift+A	Answer
Ctrl+Shift+H	Decline or hang up
Ctrl+Shift+T	Move call audio between phone and computer
Ctrl+K	Contacts
Ctrl+L	Call log
Ctrl+Shift+P	Hold or resume the call
Ctrl+Shift+M	Microphones on or off for the caller
Ctrl+1 to Ctrl+9	Input 1 to 9: heard by the caller on or off
Alt+1 to Alt+9	Input 1 to 9: mute on or off
Ctrl+I	Add an input
Ctrl+comma	Preferences
Ctrl+plus, Ctrl+minus, Ctrl+0	Larger, smaller, normal text
F10	Main menu
Ctrl+Q	Quit

Everything else works with Tab, Shift+Tab, the arrow keys, Enter, Space and Esc.

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10. Accessibility and language

- Every control has a name for screen readers, and important events are read out: incoming calls, call connected and ended, recording started and stopped, pairing codes.
- Nothing depends on colour alone: switches show text and icons, meters show numbers.
- Text size and language are in Preferences (see above).
- The full statement, in English and Norwegian, is in ACCESSIBILITY.md.

11. Troubleshooting

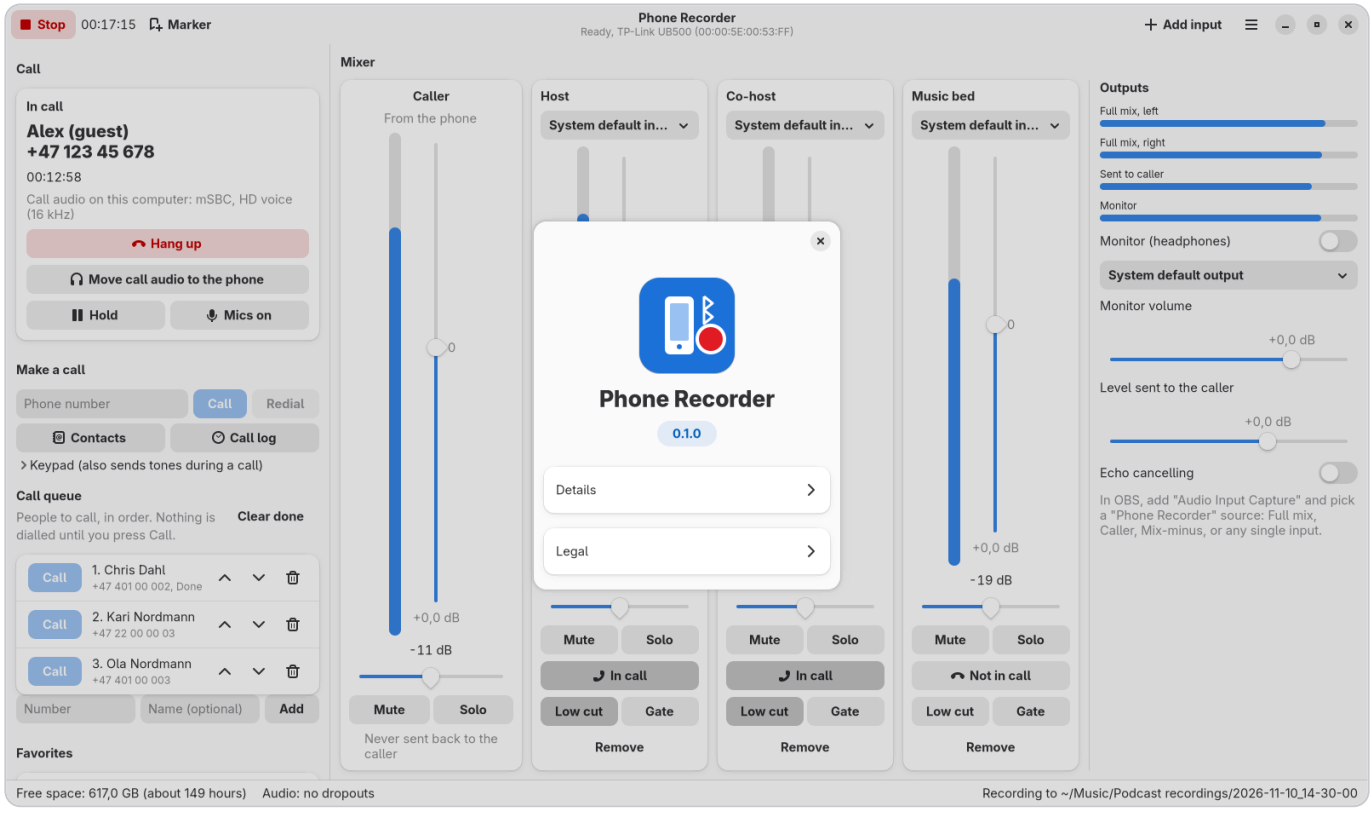
Problem	What to do
The phone does not find the app	Turn on **Pairing mode**, then search again on the phone.
"The chosen Bluetooth adapter was not found"	Plug the adapter in, or choose another one in **Choose Bluetooth adapter**.
"Adapter failed"	Save an adapter report (**Menu, Save adapter report**) and try the other mode for the adapter, or another adapter.
A call connects but no audio arrives	The app says so. In kernel driver mode, try Direct USB. Otherwise try another adapter.
Only narrowband (8 kHz)	The adapter or the phone has no wideband support. Try another adapter.
The caller hears an echo	Use headphones, or turn on **Echo cancelling**.
No contacts	Android: allow contact sharing for Phone Recorder in the phone's Bluetooth settings. iPhone: Settings, Bluetooth, (i), turn on **Sync Contacts**. Then choose refresh in the Contacts window.
Callers show only a number	The number is not in the phone's address book, or contacts are not downloaded (see above).
Clicks or dropouts	The status line counts them. Close heavy programs; check the disk is not full.
Bluetooth is still gone from the system after the app closed (rare)	Run `bt-phone-recorder --release-adapters`.

For a detailed problem report:

1. Turn on **Preferences, Bluetooth, Save Bluetooth trace** and restart the app.
2. Reproduce the problem.
3. Use **Menu, Save adapter report**, and attach it together with the newest `.pklg` file from `~/.local/state/bt-phone-recorder/` (it opens in Wireshark).

12. Where files are stored

Path	Contents
`~/Music/Podcast recordings/`	your recordings (changeable)
`~/.config/bt-phone-recorder/`	settings and remembered phone names
`~/.local/share/bt-phone-recorder/`	pairing keys, the downloaded address book, favorites, call queue and call log (readable only by you)
`~/.local/state/bt-phone-recorder/`	log file and Bluetooth traces
`~/.cache/bt-phone-recorder/`	unpacked Bluetooth firmware (recreated when needed)



Bluetooth Phone Recorder 0.1.0. https://github.com/frebergguru/bluetooth-phone-recorder